

Syllabus and also Content of the Book

IV SEMESTER

BASESF281: ENVIRONMENTAL STUDIES

(ELECTIVE FOUNDATION COURSE - 2021-22)

Credit:1
Marks : 50

Hours of Instruction / work: 2
Total hours of Instruction: 28

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- 1.2 Ecosystems - Concept, Structure and Function, Food Ecosystem
- 1.3 Bio-geographical Classification of India.
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UNIT 1

INTRODUCTION

- 1.1 Environmental Studies - Importance and Scope
- 1.2 Ecosystems - Concept, Structure and Function, Food Ecosystem
- 1.3 Bio-geographical Classification of India.
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1.1 ENVIRONMENTAL STUDIES - IMPORTANCE AND SCOPE

Living organisms do not live in isolation. They interact with each other. So there is a close association and interdependence between the living components (biotic components) and the non living components (biotic components) of the biosphere. It means that organisms live in an "Environment".

Environment consists of surrounding of an organism which includes *abiotic* and *biotic* environment. Thus, environment refers to "Sum of all i.e. water air and land, along with their inter-relationships among themselves and also other living organisms".

The word *environment* is derived from the French word "*environner*", which means "*Surrounding*". Thus, Environment can be defined as the combined interaction of physical chemical and biological conditions affecting an individual or community in the complex of social and cultural conditions.

A study of such inter-relationships between organisms and their environment is "*Ecology*".

The term ecology (oekologie) is derived from two Greek words - "*oikos*" means 'house' or place to live and "*logos*" means "*Study of Science of*". So, literally Ecology is the "study of organisms at home, in their native environment".

The term Ecology was first introduced by Reiter in 1868.

But German biologist Ernst Haeckel (1869) fully defined this term and extensively used the term Ecology in his writings. So, Haeckel popularised the term Ecology (C.B. Knight 1965)



Eugene P. Odum (1969), an American ecologist defines ecology as "the study of structure and functions of nature" and more technically as "Study of structure and functions of ecosystems".



Conventionally, two levels of approach are used in environmental studies. They are: a) Autecology and b) Synecology.

- a) *Autecology* deals with the study of the individual organism or an individual species in relation to the environment (i.e. ecology of individuals)
- b) *Synecology* deals with the study of groups of organisms which are associated together as a unit.

Man is himself an organism within an environment. Man like other animal is influenced by environment and is dependent upon other species for his needs such as food, clothing etc.

Therefore, the basic laws of ecology apply well to man like all other organisms and fundamental knowledge about environment is a must for man for his own existence on this planet Earth.

And it's a known fact that man almost has a *modifying influence* on the environment and many a times without proper regulation he has brought about *destructive effect* on the environment.

For example, man has obtained huge economic gains by using ecological principles to the fields such as agriculture, pest control, forestry, horticulture, fishery biology etc. He also has been responsible for conservation of resources such as soil, forest, wild life, water supply etc.

Even though, man has been known to control his environment successfully to meet his needs, but indiscriminate activities have resulted in short and long term undesirable ecological effects. Some of these include;

a) the introduction of different chemical poisons/pesticides that kill useful bacteria, fungi, insect along with the harmful ones.

b) The release of massive quantities of radioactive wastes, xenobiotics¹ and discarded chemicals and industrial wastes into water bodies such as rivers, sea etc resulted in pollution of environment.

1. Xenobiotics are the chemical compounds synthesized by humans which are not naturally found in living organisms and cannot be broken down (Metabolized) by them.

c) Along with the above, rapid growth of Urbanization and rapid increase of human population have been responsible for scarcity of food, open space, wild life and of survival

We, humans continue to disrupt the *energy balance* and future of human life on earth demands more *knowledge* about the ecosystems and other ecological problems.

This has made, man make *significant progress* in many areas of environmental studies

Man also has began to create legal frameworks for managing the environment. This is providing a new basis to deal with various environmental issues.

Man is himself an organism with in an environment. He is influenced by the physical aspects of his environment. He is dependent upon his environment and other species for survival.

Through his intelligence, man always has a *modifying influence* on the environment. But in his zeal to improve his standard of living, man often has destructive effects on the environment. Some of these are as follows.

- Human activities are polluting all three medium of environment i.e., air, water and soil.
- Industrial pollution, unplanned urbanization and commercial exploitation of environment and life forms is threatening the vital and fragile balance of the environment.
- Rapid rate of multiplication of human population and the growing population demands have resulted in fatal threat of scarcity of wide life, food, open space, and of survival.

Thus, human greed, ignorance and mismanagement of environment has landed the mankind at the thresh hold of an *environmental crisis*.

To solve this crisis and to preserve the components of environment for *now* and for *future*, knowledge of functional aspects of environment and the organism is desired. This desire, has lead to the development of concept called *environmental studies*.

Definition

Study of functional intricacies of environment in relation to an organism can be called as environmental study.

Scope and Importance of Environmental Studies

- Environmental studies deals with the study of fundamentals of environmental and social sciences.
- Environmental studies provides an holistic view of the environment and human - environment relationship.
- It gives an insight into the understanding of mechanisms of delicate, intricate inter relationship between organisms and environment.
- It provides an opportunity to understand major environmental issues and to arrive at possible solutions for the same.
- Environmental studies gives an insight into the anthropogenic problems of environment and ways to tackle it.
- Environmental studies provides more knowledge about ecosystems and other ecological problems.

This assumes importance as future of human life on earth demands more knowledge about the ecosystems and other ecological problems.

Multi Disciplinary Nature of Environmental Studies

For the understanding of the working of the environment, knowledge from wide ranging fields are required. Thus, environmental studies is a multi disciplinary science which depends on a variety of disciplines such as *physics, zoology, botany, chemistry, statistics, sociology, economics, mathematics, geography, demography, physiology, agricultural science, forestry, genetics, meteorology, etc.*

All these disciplines help in the better understanding of the ecological situations / principles.

According to Ernst Haeckel, Ecology can be defined as the 'scientific study of inter relationship between organisms (plants, animals) and their environment'.

Eugene P. Odum, American biologist defined ecology as the 'study of structure and functions of an ecosystem'.

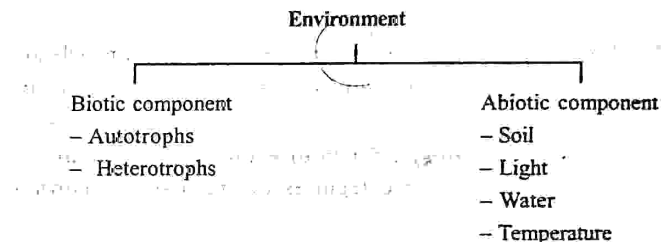
According to Charles H. Southwick, 'ecology is the scientific study of relationships of living organisms with each other and with their environment'.

Ecology is also known as Environmental biology or Environmental Science.

Environment

According to S.C. Kendeigh, Environment is the sum total of physical and biotic conditions influencing the responses of the organism.

The term Environment includes all that surrounds an organism. The organism is surrounded by the living and the non-living components. Thus, the environment has two components.



The living organisms that are present surrounding an organism is called the Biotic component. Autotrophs¹ and heterotrophs² are the biotic components present in an environment.

The non-living components namely soil, light temperature and water forms the abiotic components of an environment.

Subdivisions of Ecology

The study of ecology includes a wide variety of animals, plants and their environment. Earlier ecologists had subdivided ecology into

1. Organisms that can synthesise food by using solar energy, water and carbon dioxide.
2. The organisms that are unable to produce their food and hence dependent on autotrophs directly or indirectly for survival.

plant ecology and animal ecology. This subdivision doesn't consider the interaction between plants and animals. Moderns subdivision of ecology considers this interdependent, inseparable interrelation between plants and animals. Ecology now is often divided into **autecology** and **synecology**.

Autecology deals with the study of an individual organism and its relationship with its environment.

Synecology deals with the study of group of organism and their relationship to the environment.

Further, ecology has specialized branches like

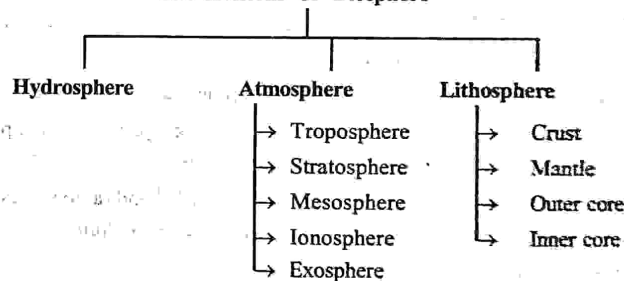
1. **Habitat ecology** : It deals with the study of different types of terrestrial, aquatic habitat.
2. **Community ecology** : Study of composition of different organisms, their dependence and interaction with one another is community ecology.
3. **Population ecology** : Growth rate and structure of populations of different organisms of the community study is called population ecology.
4. **Taxonomic ecology** : Study of ecological interactions of different taxonomic categories of organism is known as taxonomic ecology.
5. **Paleoecology** : Study of ancient environment is called Paleaecology.
6. **Ethology** : Study of animal behaviour in their natural conditions is ethology.
7. **Pedology** : This branch deals with the study of soil formation, types of soil.
8. **Applied ecology** : The branch of ecology deals with resource conservation, pollution control, land use, wild life management and agricultural forestry.
9. **Ecological energetics** : The study of food production and energy flow pattern of an ecological system is ecological energetics.

Ecological Units

In order to understand ecology, the life on earth is divided into units called ecological units. Following are the four levels of ecological units, i.e. a) **population** b) **community** c) **Ecosystem** and d) **Biosphere**.

- a) **Population** : All the individuals of a particular type of organism present in a specific area at a specific time is called population.
- b) **Community** : Assemblage of all the populations of organisms occupying a given area at a given time is community. **Karl Mobius**, German ecologist used the term **biocoenosis** to refer community.
- c) **Ecosystem** : The functional unit of ecology is Ecosystem. It includes the living organisms and their abiotic environment. The biotic and the abiotic components of an ecosystem are inter related and influence each other.
- d) **Biosphere** : The area of the earth that contains and supports life is biosphere. It includes the soil or the lithosphere, fresh and salt water or the hydrosphere and the air surrounding the earth or the atmosphere.

Sub-divisions of Biosphere



- **Troposphere** - It is the lowest region of atmosphere. Cloud formation, thunder, lightening, thunder storm formation takes place in this region. Temperature gradually decrease with height at the rate of 6.5°C per km.
- **Stratosphere** - Next to troposphere, Stratosphere is present. Clouds are absent in this region. Ozone formation occurs in this region. Temperature increases upto 90°C due to ozone formation.

- **Mesosphere** - It is the third zone of the atmosphere. In mesosphere, temperature decreases upto 80°C .
- **Ionosphere** - The atmosphere above mesosphere upto the height of about 300 Km above earth's surface is ionosphere.
- **Exosphere** - The final frontier of atmosphere is exosphere. Outer space begins from here.

Core is the central fluid or vapourised sphere with a diameter of about 2500 Kms from the centre.

From the core for about 2900 Kms above in molten state mantle is present.

Crust is the solid zone above the mantle. The surface of the crust is covered with soil.

Components of Environment

The environment consists of four segments namely;

- a) Atmosphere, b) Hydrosphere, c) Lithosphere & d) Biosphere

Atmosphere

The atmosphere consists of protective blanket of gases surrounding the earth. It is responsible for

- Sustenance of life on earth
- Saving earth from the hostile environment of outer space.
- Absorbing most of the cosmic rays from outer space and major portion of electromagnetic radiation from sun and
- Transmits only UV, visible infra red radiation and radio waves while filtering out tissue damaging UV waves below 300nm

Hydrosphere

The hydrosphere comprises all types of water resources such as oceans, seas, lakes, rivers, streams, reservoirs, polar ice caps, glaciers.

Lithosphere

Lithosphere is the outer mantle of the solid earth. It consists of minerals occurring in the earth's crust and the soil.

Eg:- Minerals, organic matter, air and water

Biosphere

Biosphere indicates the realm of living organisms & their interactions with environment, viz atmosphere, hydrosphere and lithosphere

Elements of Environment

Environment consist of following there elements. They are

- Physical Elements
- Biological elements
- Cultural Elements

- **Physical elements** of environment consists of space, land forms, water bodies, soil, rocks & minerals.
- **Biological elements** of environment includes plants, animals, micro-organisms present in Biosphere
- **Cultural elements** of environment comprises of economic, social and political elements and are essentially features made by man.

Importance of Environmental studies

Environmental studies enlighten us about the importance of protection and conservation, and also indiscriminated release of pollutants into the environment.

At present a great number of environmental issues have grown in complexity and size threatening the very survival of mankind on earth. This makes it of paramount importance to create awareness among the population of humans leading to safe guarding of components of ecosystem.

Scope of Environmental Studies

Environmental studies is found to have multiple and multilevel scopes. Following is the listing of the scopes of environmental studies.

- It provides the knowledge about ecological systems and cause and effect relationships among the various components of ecosystem.
- It provides information about the biodiversity and possible dangers to the flora and fauna of the environment.
- It crates ample awareness among the people about various renewable and non renewable resources of the region.

4. It leads to the analysis of use of various resources available & its sustainable use.
5. It brings about the reasons, and consequences caused due to natural disasters and manmade disasters such as landslides, floods etc. and pollution.
6. The environmental studies enable an citizen who is "environmentally literate" by knowing about various environmental acts, rights, rules, regulations etc to take appropriate decisions for the protection and improvement of the environment.
7. It gives an insight into the problems of over population, aspects associated with maintenance of health and hygiene. It also explains how arts, science and technological advances can help in removing the ill effects of the above factors from our society.
8. The study leads to the development of various techniques that are eco-friendly in tackling various environmental issues.
9. It enables the citizens learn to use the resources judiciously so that the resources, we inherited from our ancestors are passed on to our future generation in pristine form.
10. It gives in-depth knowledge about environment leading to the use of components of environment without causing any deterioration.

1.2 ECOSYSTEMS: CONCEPT, STRUCTURE AND FUNCTIONS; POND ECOSYSTEM

1.2.1 Ecosystem: The concept

The ecosystem/ecological system is the "Functional unit" of ecology.

Ecosystem may be defined as "any natural unit that includes a community of organism and their complex environmental factors, interacting to produce a stable system."¹

Examples of ecosystem includes lake, pond, estuary forest etc.

The largest biological system where several ecosystems operate together is known as "Biosphere" or "Ecosphere". The term "Biosphere" was first used by an Austrian geologist Eduard Suess (1875). The concept

1. Community consists of assemblage of all the populations of organisms occupying a given area at a given time.

was further developed by Russian mineralogist Vernadsky (1926) and Hutchinson (1970) defined Vernadsky (1926) and Hutchinson (1970) defined biosphere as "that part of the earth in which life exists."

Lindeman (1942) defined ecosystem as "a system composed of physical-chemical-biological processes active within a space-time unit of any magnitude"

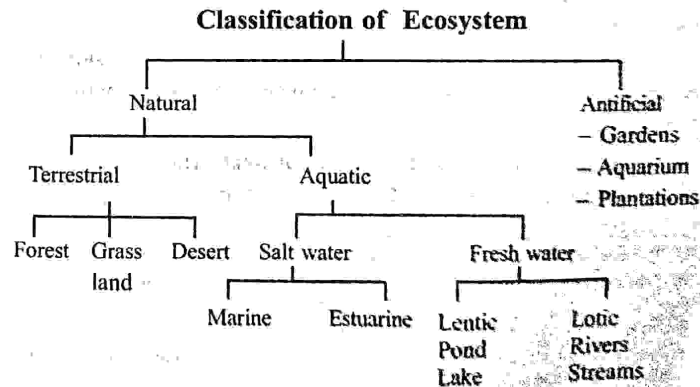
Odum (1963) defined ecosystem as "the basic functional unit of nature including both living (biotic) organisms their non living (abiotic) environment, each interacting and influencing each others properties and both being essential for the development and maintenance of the system".

Depending upon various biomes, ecosystem can be divided into following groups;

1. **Mega ecosystems**:- These include oceans, seas, estuaries and brackish waters, rivers, forests, grasslands and deserts
2. **Meso ecosystems**:- These consists of deciduous forests and coniferous forests.
3. **Micro ecosystems**:- It is comprised of croplands, cities, towns, villages (artificial) and ponds, lakes and rivers (natural).

Classification of Ecosystem

Primarily ecosystems can be broadly classified into two major divisions : natural and artificial.



Ecosystems as occurring in nature are natural ecosystems. Depending on the habitat,¹ natural ecosystems are classified into (1) *Terrestrial or land based ecosystems* and (2) *Aquatic or water based ecosystems*.

Ecosystems constructed by human beings like gardens, plantations or aquariums are called *artificial or agro ecosystems*.

1. Terrestrial Ecosystem

Land based ecosystems are known as terrestrial ecosystems. Three major types of terrestrial ecosystems are present : a) *Forest ecosystem* b) *Grass land ecosystem* and c) *Desert ecosystem*.

2. Aquatic Ecosystem

Water based ecosystems are called aquatic ecosystem. Based on the salt content of water, aquatic ecosystem is classified into two types namely a) The salt water ecosystem b) The fresh water ecosystem

1.2.2 ECOSYSTEMS - STRUCTURE AND FUNCTION

Structure of an Ecosystem

In an ecosystem, there are two major interacting structural components present : a) Abiotic component and b) Biotic component

A) Abiotic Component

It comprises of non living component of the environment. Light, soil, air and water are the abiotic components.

The energy of sunlight is essential for the basic physiological processes such as photosynthesis,² transpiration,³ germination and flowering.

Physical and chemical processes of weathering⁴ is responsible for the formation of the soil. Microscopic and macroscopic plants and

1. Environment in which an organisms live.

2. Process of formation of food by organisms using sunlight, water and carbon dioxide.

3. evaporation of water from the aerial parts of the plants.

4. process of break down of bigger rock particles into smaller mineral particles with water and air in the spaces.

animals of wide variety reside in the soil playing an important role in the functioning of the ecosystem.

Air is a mixture of nitrogen, oxygen, carbon dioxide and traces of inert gases. About 78% by volume of nitrogen is present in the atmosphere. Where as oxygen and carbon dioxide are present in about 20% and 0.032% by volume respectively. Oxygen is vital for the survival of the organisms. Carbondioxide is one of the raw material for photosynthesis.

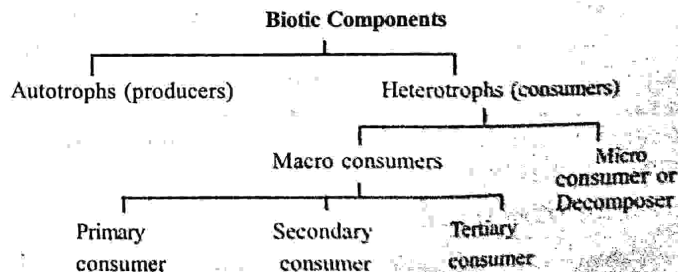
Water covers 79% of earth's surface and is the medium for aquatic life. Growth of the plants are dependent on the availability of water. Plants and animals living in water are called as aquatic organisms. Aquatic organisms are of two types - fresh water and marine.

The major component of the protoplasm of the cell is water. It is essential for the metabolic reactions occurring in the organisms.

Besides the above, the climatic factor like temperature gases dissolved in water such as oxygen and carbon dioxide etc, salts of phosphorous, calcium etc and organic compounds such as amino acids, carbohydrates etc. form the important abiotic constituents.

B) Biotic Component

It consists of living components of the environment. Microscopic and macroscopic plants and animals form the biotic components of an ecosystem. These are grouped into two categories - **Autotrophs** and **Heterotrophs**.



Autotrophs : Autotrophs are organisms that are capable of producing the food using the green pigment chlorophyll from sunlight, carbon dioxide and water. Since food is produced by autotrophs, they are also called as the producers. The energy needed for the existence of ecosystem is produced by the autotrophs. Large trees, grasses, rooted and floating plants, the microscopic algae and diatoms are some of the examples of autotrophs.

Heterotrophs : Heterotrophs are organisms who are unable to produce the food as they do not have the chlorophyll. For survival, they are dependent on the autotrophs *directly* or *indirectly*. Hence, the heterotrophs are known as the **consumers**.

The consumers are of two types (a) Macroconsumers or phagotrophs and (b) Micro consumers /Decomposers or Osmotrophs.

a) Macro consumers : They are large sized consumers with a digestive system capable of enzymatically digesting larger food molecules into smaller food molecules. The macro consumers are further grouped into three types (i) Primary consumers (ii) Secondary consumers and (iii) Tertiary consumers.

(i) Primary consumers : A primary consumer is directly dependent on the producer for food. These are also known as **herbivores**.

(ii) Secondary consumer : These are the organisms that feed on primary consumer. A secondary consumer is also called a **primary carnivore**.

(iii) Tertiary consumer : These are the organisms, that obtain their food by feeding on either primary or secondary consumers.

b) Micro consumers / Decomposers : These are microscopic bacteria and fungi. They decompose excreta of animals and dead parts of plants and dead bodies of animals. They are also called **detritivores** - dirt eaters. Sometimes they are referred to as the **natural scavengers** of the ecosystem.

FUNCTION OF AN ECOSYSTEM

Functioning of an ecosystem is studied with reference to two aspects. They are;

1. **The flow of energy** is various directions, from and to within the living organisms, at different trophic levels.

2. **A cyclical passage** known as biogeochemical cycle in which materials such as carbon, nitrogen, oxygen, sulphur, phosphorus and water circulate between the living and the non living components of an ecosystem.

I Energy flow

The main sources of energy for an ecosystem is the light energy derived from the sun (radiation energy). The passage of energy in an ecosystem comprises of following six processes. They are;

1. The autotrophs (green plants) convert the radiant energy of the sun into **chemical energy** by photosynthesis and energy is stored in the bonds of the sugar molecules.¹
2. A part of this chemical energy is used for metabolic activities like growth, reproduction etc. During this process, some amount of energy is lost as heat.
3. The autotrophs become the food and energy source for the **herbivores/primary consumers** for their metabolic activities. Energy is again lost as heat in the form of compounds such as urea & uric acid, which still contain usable energy.
4. The carnivores (Secondary consumers) consume herbivores and use them as source of energy with the same end result as in (3)
5. The tertiary consumers consume/eat secondary consumers for the same purpose and with the same result as in (3)
6. Decomposers degrade dead bodies of plants and animals and also the waste products of animals. This process is called **decomposition** and it consists of several steps. The end products of decomposition are without energy. These are returned to the **abiotic environment** and are again used by plants as nutrients.

As a result of the above six processes, the energy passes continuously in an ecosystem and matter is recycled again and again.

1. Photosynthesis may be defined as an anabolic (constructive/building) up process in which carbohydrates are synthesized from carbon dioxide and water by green plants using sunlight and chlorophyll present in the chloroplast. Oxygen is produced as a by product.

At each step, between the various levels of consumers energy is lost as heat. Hence, the transfer of energy is not 100% efficient and some energy is always wasted.

Trophic level

The flow of energy at successive stages in an ecosystem is known as "trophic level". In ecology, "trophic" relates to "feeding and Nutrition". It is derived from the Greek words trophe and trephein meaning food and to feed respectively. Trophic level comprises of a group of organisms in an ecosystem which obtain their energy from eating grass such as cattle, grasshoppers etc. are part of the same trophic level.

Aggregation or grouping of different trophic levels within an ecosystem is known as "trophic structure" of an ecosystem.

1) Typically an ecosystem has 3 to 4 trophic levels through which energy flows. In other words, there are usually 3 to 5 groups of organisms which derive their nutrition in the same general way. i.e., 1. The autotrophs/producers-P occupy the first trophic level-T₁

2) The herbivores are the heterotrophs.¹ These are the primary consumer-C₁, that obtain their energy directly from the autotrophs, and occupy the second trophic level.-T₂.

3) The carnivores (meat eaters) or secondary consumers-C₂ that obtain their energy indirectly, from the autotrophs, occupy the third trophic level-T₃.

Examples		
Tertiary Consumer - C ₃	Fourth Trophic level (Top carnivore) - T ₄	Man, Lion etc
Secondary consumer - C ₂	Third Trophic Level (Carnivore) - T ₃	Birds, Fishes, Ass etc.
Primary Consumer - C ₁	Second Trophic level (Herbivores) - T ₂	Cow, Grasshoppers, Zooplankton etc
Primary Producer - P	First Trophic Level (Plants) - T ₁	Grass, Trees, Phytoplankton

Diagrammatic Representation of trophic levels in an ecosystem

1. heterotrophs are the organisms which cannot produce their own food and are dependent on autotrophs for their energy need.

Thus, the organisms of each trophic level obtain their nutrition from those in the level below. As energy transformation is not efficient, animals at successive trophic levels obtain lesser amount of energy.

Food Chain

The trophic structure of an ecosystem is the basis of the food chain. Food chain is a system comparable to a chain with series of links in it. Thus, food chain is the sequence of organisms in which one organism feeds on that preceding it.

Example:-

Grass → Grass hopper → Frog → Eagle.

Explanation:

- 1) Producers (Grass) is always the first link in a food chain. These produce the food by using sunlight.
- 2) Herbivores (Grass hopper) that feeds on plant or plant parts is the second link in the food chain.
- 3) The third link in the food chain is the secondary consumer or a carnivore (Frog)
- 4) The fourth link in the food chain is the tertiary consumer which is also a carnivore.

Basically, food chain is of two types they are;

a) Grazing food chain (GFC)

And

b) Detritus food chain (DFC)

a) The grazing food chain begins with green plants, (producers) goes to grazing herbivores (Primary consumers) and to carnivores (Secondary or tertiary consumers)

i.e, Grass → Goat → lion
(Producers) (Primary consumer) (Secondary consumer)

Here, the energy is transferred from Green plants to primary and secondary consumers.

b) The detritus food chain begins with decaying organic matter of plant and animal bodies called as *detritus*. The food energy is transferred

from detritus to detritivores (Earthworms, Millipedes etc) Detritivores, in turn are fed upon by larger organisms building up a food chain.

i.e.

Detritus → Earthworms → Frogs → Snake
(Producer) (Primary consumer) (Secondary consumer) (Tertiary consumer)

BIOGEOCHEMICAL CYCLES

Biogeochemical cycles are the cyclical passage of biogeochemical like carbon, hydrogen, nitrogen, phosphorus and sulphur between the living component(bio) and the non living component, the earth(geo) of the biosphere so that what is provided by earth is returned back to it.

Broadly, there are two types of biogeochemical cycles. They are;

1) Gaseous cycle

Eg:- Carbon cycle, Nitrogen cycle etc

2) Sedimentary cycle

Eg:- Phosphorus cycle, sulphur, cycle etc.

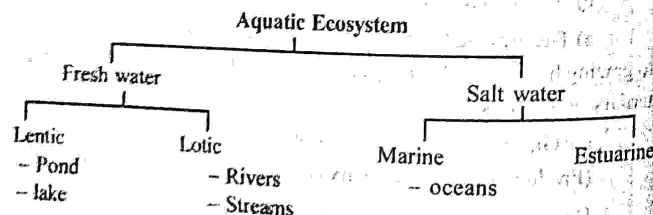
1.2.3 POND ECOSYSTEM

Pond Ecosystem is an example for aquatic ecosystem

Aquatic Ecosystems

Water covers about three quarters of the earth's surface. This is either in the form of oceans or fresh water. Thus the aquatic ecosystems are broadly classified into two types;

A) Fresh water ecosystem and B) Saltwater ecosystem



Fresh water ecosystems are divided into two groups based on the nature of fresh water - Lentic and Lotic. In lentic freshwater ecosystem, the water is standing still. Ponds, lakes, bogs, swamps etc are examples of

lentic. In the lotic, the water is running in nature i.e it is in continuous motion. Rivers, Streams are examples of lotic.

The salt water ecosystem includes the marine and estuarine habitats. Estuary represents transitional zone between a river and the sea. It has dissolved contents intermediate between those of fresh water and salt water.

Lentic ecosystems - All standing fresh water habitats such as ponds, lakes, bogs, swamps etc. are examples of lentic ecosystem.

Inland depressions containing standing water are called lakes. They vary in size from small ponds of less than a hectare to large water areas covering thousands of square kilometer. The depth varies from a few centimeters to over 1600 meters.

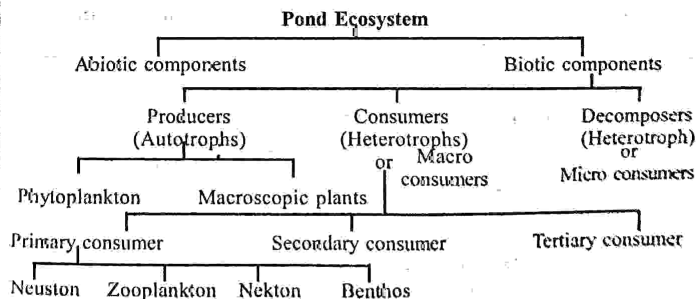
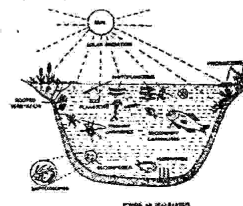
Ponds are small bodies of standing water. In relation to light intensity, vertical stratification is seen in ponds and lakes. There are three well-recognised strata namely :

1. **Littoral zone** - It is the shallow water near the shore.
2. **Limnetic zone** - It is open water zone away from the shore
3. **Profundal zone** - It is the deep-water area below the limnetic zone

Besides these, an abyssal zone is present in deep lakes. It begins at about 2,000 meters from the surface.

Structural and Functional Components of Pond Ecosystem

The pond ecosystem has two major interacting components namely Abiotic components and Biotic components.



Abiotic Components-It is also known as the non-living component. The abiotic components of pond ecosystem includes :

1. **Light** - Sunlight is the most important abiotic component. It provides the energy needed for the photosynthesis leading to the production of food by the producers.

2. **Water** - Water forms the habitat of the organisms. Water also contains dissolved salts of Calcium, Potassium, Sodium etc. Dissolved gases like oxygen, carbon dioxide, nitrogen etc are also present in water.

Apart from these, the climatic factors like temperature is also an important abiotic factor.

Biotic Component

It is also known as the living components of the ecosystem. They are grouped into two categories i.e I. **Autotrophs** and II. **Heterotrophs**.

I. **Autotrophs** are also known as the **producers** of the ecosystem. They use sunlight, water, carbon dioxide and green pigment chlorophyll to produce the food. Autotrophs forms the first trophic level of the ecosystem. There are two types of autotrophs present in pond ecosystem.

a) **Phytoplankton** : Minute floating plants such as algae and diatoms present throughout the area of pond are called phytoplankton. Eg : Blue-green algae, Green algae, Diatom etc.

b) **Macroscopic plants** : These are the large water plants also known as hydrophytes. It consists of emergent vegetation¹ submerged vegetation² & free floating vegetation. Eg : - Sagittaria, Jussiea, Hydrilla, Pistia, Vallisneria etc.

II. **Heterotrophs** are the animals that cannot produce their own food. They are dependent on producers for the food energy. They are of two types :

1. Consumers and 2. Decomposers

1. **Consumers** : Consumers are relatively large sized with a digestive system. These are further grouped into following types : a) **Primary consumers** b) **Secondary consumers** and c) **Tertiary consumers**

1. Plants rooted in the shallow regions of the bank. Leaves emerge out of water.
2. Plants completely submerged under water.

a) **Primary consumers** : These depend directly on producers for food. Primary consumers form the trophic level II. Primary consumers are of following types :

(i) **Neuston** - Unattached organisms that live at the air-water interface are called Neuston. It contains floating plants like duckweed and animals such as water striders. Insects such as diving beetles, larvae, tadpole etc. also are other examples of neuston.

(ii) **Zooplankton** - The animals that drift on water surface controlled by the water current are called Zooplankton. Eg : - Cyclops, Daphnia etc

(iii) **Nekton** - These are the aquatic swimming animals. They can swim independent of wave and water current as they possess locomotory organs. Eg : - Insect larvae, Insects etc.

(iv) **Benthos** - These are the organisms living at the bottom of the pond. They may be either found living in the sediment or found at the interface of water - sediment. Eg : - Crustaceans, Molluscs etc.

b) **Secondary Consumers** : These are the organisms that feed on the primary consumer. Secondary consumers form the trophic level III. It is also called as a primary carnivore.

Eg : Water scorpion (Ranatra)

Water bug (Belostoma)

Fishes etc.

c) **Tertiary Consumers** : The animals that feed on the secondary consumers are called tertiary consumers. The tertiary consumers forms the trophic level IV. Eg : - Cat fish, water snakes etc

2. **Decomposers or Microconsumers** : These form the trophic level V of the pond ecosystem. It consists of aquatic bacteria and fungi. They decompose the dead bodies and excreta of the consumers. This helps in recycling of minerals in the pond ecosystem. This recycling is essential for the stability of the ecosystem.

FUNCTIONAL ASPECT OF POND ECOSYSTEM

Producers by photosynthesis synthesise food. This energy present in the producers is passed on sequentially from primary, secondary to tertiary consumers. Decomposers are responsible for the recycling of the minerals in the ecosystem. Thus all the different components of the ecosystem interact and influence one another.

1.3 BIO-GEOGRAPHICAL CLASSIFICATION OF INDIA

Introduction

Bio-geography is the study of the distribution of species, communities and ecosystems in geographic space and through geological time. Biogeography includes the area, soil, water, climate as well as plants and animals of particular geographical region. Plants contribute much towards the biogeography than animals as plants are stable/non movable.

Biogeographical classification deals with the biodiversity found in different parts of the world. Plants are considered to be more important in the biogeographical classification as they are non movable.

Alfred Russell Wallace, has classified the entire world into six biogeographically regions. They are;

- 1) Neoaectic
- 2) Paleoaectic
- 3) Ethiopian
- 4) Australian
- 5) Neotropical and
- 6) Oriental

India is included in oriental region along with Sri Lanka, Philippines, Sumatra, Java, Bali and some parts of China.

Special Features of Oriental Region

1. Most of the area is covered with thick forest
2. Advanced type of fishes are seen in this region
3. Most of the region has monsoon climate.

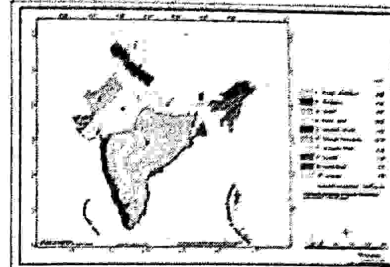
BIO-GEOGRAPHICAL CLASSIFICATION OF INDIA

India is the seventh largest country in the world and second largest nation in Asia with an area of 3,287,263 sq. km exhibiting wide range of landscape that are rich in natural resources.

Biogeographers have classified India into Ten Bio-geographic zones. This classification is based on geography climate, soil, pattern of vegetation and communities of invertebrates Amphibians, insects, Reptiles, Birds and mammals.

The ten bio-geographic zones of India are as follows;

1. Trans Himalayan zone
2. Himalayan zone
3. Desert zone
4. Semiarid zone
5. Western Ghat zone
6. Deccan Plateau zone
7. Gangetic Plain zone
8. North-East zone
9. Coastal zone
10. Islands



Bio-geographic Features

1. Trans-Himalayan Zone

The Himalayas consists of the youngest and loftiest mountain chains in the world. The Himalayas have attained a unique feature of its own due to its high altitude, steep gradient and rich temperate flora. The Himalayas are extended to the Tibetan plateau.

The Himalayan region present to the North of the Great Himalayan range are called the trans-Himalayas. It accounts for 5.7% of the India's land mass. The high altitude cold deserts of Ladakh and Lahaul Spiti of Himachal Pradesh are found in this region.

The trans-Himalayan region has very little vegetation. It has the wild sheep and goat community living in this region. Snow leopards are also found. Along with these, migratory black-necked crane is found.

2. Himalayan Zone

The Himalayas form the northern boundaries of India. The entire mountain chain is running from Kashmir in the North-west to Assam in the North-east. The Himalayas cover 7.2% of India's land mass and harbors diverse range of biotic provinces and biomass.

The forests of this zone are very dense with tall evergreen trees and extensive growth of grass. Oak, chestnut, conifer, pine and deodar are the common types of trees found in this region. Vegetation is not found above the snow line.

Several types of animals live in this region. They include wild sheep, mountain goats, ibex shrew and Tapir. Snow leopard and panda are also found.

3. Desert Zone

The extremely dry area west of the Aravali Hill range consists of

- (i) salty desert of Gujarat and
- (ii) Sandy desert of Rajasthan

Desert zone occupy around 6.9% of India's land mass.

The types of desert found in the India are;

- a) The desert of western Rajasthan
- b) The desert of Gujarat
- c) High-altitude cold desert of Jammu, Kashmir and Himachal Pradesh.

The Indian desert has more diversified fauna. The plants grown here are most xerophytic in nature. Babul, kikar, wild palm grows in these areas.

The endangered bird Bustard is found here. Camels, wild asses, snakes and foxes are found in hot and arid parts of the desert.

4. Semi-Arid Zone

Generally the areas adjoining to the desert are referred as semi-arid zone. In India, the transitional zone between the desert and the dense forest of western Ghats represent the semi-arid zone.

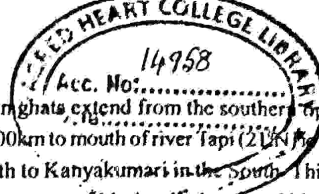
In this semi-arid zone, thorn forest is the natural vegetation. The characteristic feature of the semi-arid zone vegetation is the presence of discontinuous vegetation cover with open area of bare soil and throughout the year one can observe the deficit of soil water.

Thorny shrubs, grasses, bamboos, xerophytic herbs and ephemeral herbs are found here.

Birds, Eagles, Snakes, fox, jackals, buffaloes and leopards etc are the types of animals found in the semi-arid area.

5. Western Ghats

Western Ghats constitute one of the unique biological regions of the world. The Western Ghats are mountain range that runs along the western



coast of India. The western ghats extend from the southern tip of peninsula (8°N) northwards about 1600 km to mouth of river Tapi (23°N) from Southern Tip of Gujarat in the north to Kanyakumari in the south. This ghat section covers an extremely diverse range of biotic provinces and biomes. It covers about 5.8% of the India's land mass. The varied climate and diverse topography is responsible for wide variety of habitats which support unique sets of plants and animals. Besides this rich biological diversity the region has high levels of cultural diversity as the forests are inhabited by many indigenous people.

As a result, western Ghats are amongst the 25 biodiversity hot-spots of the world. These western Ghats are known for the high levels of endemism¹. These endemic plants are associated with the evergreen forests of the area.

6. Deccan Plateau Zone

It is a large triangular plateau south of the Narmada valley. Three sides of the plateau are covered by mountains slopes towards east; Satpura Mountains cover the north while western Ghats covers the west side and Eastern Ghats cover the eastern side of the plateau. It covers approximately 15.6% of the country's land mass. The western and Eastern Ghats meet each other at the Nilgiri hills. The western Ghat includes Sahyadri, Nilgiris, Anamalai and Cardamom Hills. Rivers like Mahanadi, Godavari, Krishna and Kaveri originate from Western Ghats and flow eastwards. The Eastern Ghats are broken into small hill ranges by the rivers coming from the Western Ghats. Most of these rivers culminate in the Bay of Bengal. The Godavari is the longest river in the Deccan plateau. The Narmada and Tapi flow westwards and reach the Arabian Sea. The entire zone is covered mostly with deciduous trees.

7. Gangetic Plain Zone

The largest unit of the great plain of India is the Gangetic plain zone. This plain covers the area between the south Himalayas to the tropic of cancer. It covers about 11% of the country's landmass. Ganga is the main river of this plain. River Ganga covers about 72,4 mha area and the river Brahmaputra forms the main drainage axes in the major portion.

1. The animals/plants found exclusively in a particular region.

The soil and sediment thickness is maximum in the plains of Ganga. The physiographic scenery varies from arid and semi arid landscapes of desert plains to the humid and pre humid landscapes of the delta and alluvial plains in the east.

The plain supports the highest population densities depending upon purely agro-based economy of these regions. Teak, Sal, Shisham, Mahua and Khair are some of the varieties of trees found in the forests of the plain.

8. North-East Zone

North-east zone consists of the plains and non-Himalayan ranges of northeastern India. It covers about 5.2% of the India land mass. It has several species of orchids, Bamboos, ferns, etc. In this zone, wild relatives of cultivated plants such as banana, mango, citrus and pepper are grown.

9. Coastal Zone

India has over 7,516.4 km of coastline. The west coast and East coast of India varies in their characteristics and structural features. The west coast is narrow with the exception of Gulf of Cambay and Gulf of Kutch. The west coast becomes a bit wider towards extreme south along the south of Sahyadri. The west coast is known for its charming scenic backwaters. In contrast to the west coast, the east coast is broader and it is due to the deposition of soil from the east flowing rivers that has led to the change in the base levels. In this coast, the characteristic feature is the formation of extensive delta of the rivers-Godavari, Krishna and Kaveri.

Along the estuarine tracts, Mangrove, vegetation is a typical characteristic feature.

Fertile soil cover is present over larger parts of the coastal plains. All along the coast line, coconut trees grow and rice is the main crop of this area.

Along with coconut, rubber is the main vegetation of the coastal area.

The main states found on the coastal area of India are - Gujarat, Maharashtra, Goa, Karnataka, Kerala, West Bengal, Odisha, Andhra Pradesh, Telangana, Tamilnadu and Pudicherry.

10. Islands

Geographically, India has two groups of islands namely the Arabian sea islands and the Bay Islands.

Both groups of islands differ significantly in origin and physical characteristics.

The Arabian Sea islands (Laccadive, mini coy etc) are founded by the remnants of the old land mass leading to coral formation. Some of the islands of this area are fringed with coral reefs and many of them are covered with thick forests.

On the other hand, the Bay Islands lay only for about 220km. The Andaman and Nicobar islands in the Bay of Bengal has about 360 big and small islands. Among these, only five islands are inhabited. Tribal people are found only on island of Nicobar. These Islands have a highly diverse set of Biomes and occupy 0.03% of India's biomass.

1.4 BIODIVERSITY

1.4.1 Biodiversity-meaning

The concept of Biodiversity first appeared in 1980. The word "Biodiversity" is shortened form of two words "Biological" and "Diversity". The term biodiversity was coined by W.G. Rosen in 1985.

Biodiversity (GK. Bios=life; diversity=forms) or BIOLOGICAL DIVERSITY can be defined as "the vast array of species of micro organisms, algal, fungi, plants and animals occurring in the earth either in the terrestrial or aquatic habitats and the ecological complexes of which they are a part".

According to the United Nations Earth Summit (1992), biodiversity is defined as "the variability among living organisms from all sources, including terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part; this includes diversity within species, between species and of ecosystem.

Biodiversity differs from place to place. This is because of environmental conditions of the area as well as ability of an organism's level of tolerance to that area.

Level of Biodiversity

Biodiversity occurs at different levels. Some of them are;

- 1) Genetic diversity
- 2) Species diversity
- 3) Ecosystem/Community diversity

1. Genetic diversity

Genetic diversity is related to the variations of genes within species.

These variations may be due to

- Variants of the same gene
- Variation in the entire gene pool
- Variation in the chromosomal structure.

Because of these genetic differences, formation of sub species, varieties, races, strains etc are observed.

Eg:- Varieties of Mangoes, wheat, Rice, etc.

2. Species diversity

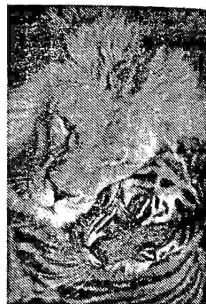
Species diversity is related to variety in the number and richness of the species within a region.

Eg:- The big cats i.e Tiger (*Panthera tigris*)

Lion (*Panthera Leo*) and

Snow leopard (*Panthera uncia*)

belong to the same genus panthera but they all differ at the species level.



3. Ecosystem/Community diversity

This is related to different types of ecosystems/ habitats. A habitat is the cumulative factor of the climate, vegetation and geography of a region.

Measurement of Biodiversity

Biodiversity is measured by two major components

- They are; a) Species richness
b) Species evenness

- a) Species richness refers to the number of species per unit area. As the area considered increases, the number of species found in it also increases due to the availability of more resources. Species richness is the measure of number of species found in a community.
- b) Species evenness refers to the relative abundance with which each species is represented in an area. It measures the proportion of species at a given site.

Eg:- Amphibian species diversity is greater in western Ghats than the Eastern Ghats.

According to conservation international (CI), 1996

1.4.2. Hotspots of Biodiversity in India

Hotspots are those regions of rich biodiversity which have been declared sensitive due to direct or indirect interference of human activities.

The concept of hotspot was first developed by Norman Meyers in 1988.

There are four main criteria for the classification of hotspot. They are

- Rich species diversity
- Number of endemic species
- Degree of threat in terms of habitat loss and
- Degree of exploitation

According to conservation International CI, 1996, a region to be called as hotspot should fulfill the following considerations. They are;

- The region should have at least 1500 species of vascular plant i.e more than 0.5% of the world's total plants.
- The region should have lost greater than or equal to 70% of its original habitat

India has always been on the list of the richest countries in the world for its biodiversity In India, there are approximately

- 350 species of mammals which is about 7.6% of world species
- 1224 species of birds accounting for 2.6% of world species
- 197 species of amphibians which make up 4.4% of world species
- 408 species of reptiles which is about 6.2% of world species
- 2546 species of fishes which makes up 11.7% of world species

- 15000 species of flowering plants which are about 6% of the world species

India is recognized as one of the mega-diverse countries, rich in biodiversity and associated traditional knowledge.

23.30% of geographical area of India is under forest and tree cover. Even though India is just 2.4% of land area of world it accounts for nearly 7% of the recorded species. In 2005 conservation international (CI) published an updated titled book "Hotspots Revisited: Earth's Biologically Richest and most Endangered Terrestrial Ecoregions" and gave the list 35 biodiversity hotspots of the world.

These 35 biodiversity hotspots cover 2.3% of the Earth's land surface, yet more than 50% of the world's plant species and 42% of all terrestrial vertebrate species are endemic to these areas.

Out of these 35 hotspots India contains four(04) hotspots. They are,

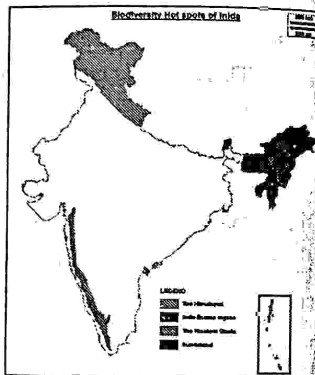
1. The Himalayas
2. Indo-Burma
3. Western Ghats and Sri Lanka
4. Sundalands

1. **The Himalayas:** It includes the entire Indian Himalayan Region (and that falling in Pakistan, Tibet, Nepal, Bhutan, China and Myanmar).

2. **Indo-Burma:** Indo-Burma region includes entire North-Eastern India, except Assam and Andaman group of islands.

3. **Western Ghats and Sri Lanka:** It includes the entire Western Ghats (and Sri Lanka).

4. **Sundalands:** Sundaland consists of Andaman and Nicobar group of islands (and Indonesia, Malaysia, Singapore, Brunei and Philippines)



1.4.3. Threats to Biodiversity

Man in guise of development has been directly or indirectly involved in the destruction of the environment. So any direct or indirect

human activity that results in destruction and becomes a threat to the survival of organisms can be called as threat to biodiversity.

A list of human activities that are acting as threats to biodiversity are listed, below; They are;

- 1) Extinction
- 2) Genetic and behavioral degradation of taxa
- 3) Habitat destruction
- 4) Habitat degradation and Fragmentation
- 5) Pollution
- 6) Global climate change
- 7) Introduced species
- 8) Disease
- 9) Exploitation

1. Extinction

The most obvious loss of biodiversity is the process of extinction. "When no more individual of a taxonomic group survive, either within a specified part of their range or forever lost across their entire range" can be called as extinction.

Throughout the history of life on this earth, five episodes of mass extinctions have occurred and sixth one is in progress. The sixth one is the fastest as it is anthropogenic i.e involvement of humans is further aiding the extinction of species at a faster rate and faster pace.

2. Genetic and behavioral degradation of Taxa

- The outright loss of population and
- Alteration of population as a result of human activities are responsible for the genetic and behavioral degradation of Taxa

Eg:- Genetic and behavioral degradation of species such as spix's macaw (*Cyanopsitta spixi*)

3. Habitat Destruction

Human Activities is becoming responsible for the outright loss of habitat resulting in devastating threats to biodiversity.

Conversion of land for *urban and agricultural use* are the main causes of habitat destruction.

Habitat destruction of terrestrial ecosystems occurs due to various phenomena such as:

- Deforestation
- Desertification
- Urbanization and
- Artificial burning etc

Habitat destruction of aquatic ecosystems occurs due to various phenomena such as:

- Dam construction
- Wetland filling
- Water flow diversion
- Oil drilling and
- Bottom trawling etc

4. Habitat degradation and Fragmentation

"When some aspect of the natural environment is removed or altered" can be called as habitat degradation.

As a result of human alteration and destruction of habitat, increase in habitat fragmentation of an ecosystem is observed. This has resulted in isolation of habitats to small fragments of their original size. This fragmentation upsets the ecosystem by decreasing the ability of species with large home range or specialist habitats to survive.

5. Pollution

"Contamination of the natural environment" is pollution

Since industrial revolution the input of organic and inorganic substances into the environment by anthropogenic (Human) activities has become a major growing threat to biodiversity.

6. Global climate change

Organisms interact with their environment and any change in environment acts as a key determinant whether an organism thrives or

becomes extinct. So, perhaps the most crucial mechanism of environmental change is global climate

Fluctuations such as slight changes in average daily temperature, duration of rainy season, night time temperature, the solar radiation etc can effect any biological organism.

Human activities that affect global climatic change includes;

- Production of air pollutants from source such as fossil fuel combustion and burning of forest.
- Agricultural activities results in reducing available forests from slash and burn agriculture and
- Release of methane-a potent green house gas from livestock.

7. Introduced species.

Introduced species are those species that are found in areas outside of their native range due to direct or indirect human activity. Once successfully introduced into new area these introduced species establish themselves. They become invaders and grow abundantly due to the absence of natural predators in that region and this subsequently results in the loss of biodiversity.

8. Diseases

A special case of threat to biodiversity caused by species introduction is the expansion of pathogens or parasite resulting from human activity. Diseases can result from genetic disorders, pathogens such as viruses, bacteria and other parasites. Coevolution of hosts and pathogens over evolutionary time results in coexistence of both host and pathogens. Due to human activity, when imbalance is produced it results in creating conditions that opens the pathogens to spread. Diseases are often transmitted across different species with the new host species often devastated by the new pathogen. This acts as threat to biodiversity.

9. Exploitation

One of the most direct anthropogenic manmade mechanisms for the loss of biodiversity is extraction of organisms from their natural environment.

Humans extract biological organisms from nature for food and other resources. Wide spread exploitation in the form of fishing, hunting and logging of forest is observed. Vertebrate species are especially targeted for food and other economic benefit.

Despite CITES (Convention on the International Trade in Endangered Species) endangered animals are still decimated by poaching.

A special class of exploitation is the logging of forests which presents probably the greatest single threat to biodiversity.

1.4.4. Endangered and Endemic species

Endangered species: Endangered species is a species that is very likely to become extinct in the near future either worldwide or in a particular jurisdiction. These species may be at risk due to factors such as

- Habitat loss
- Poaching
- Invasive species
- Diseases
- Human exploitation

Red List of International Union for Conservation of Nature (IUCN) lists the global conservation status of many species. The IUCN is the global authority on endangered species determination. The IUCN compiles information from a network of conservation organizations to rate which species are most endangered and this information is published in the IUCN Red list of Threatened species. The IUCN regional red lists assess the risk of extinction to species in over 100 countries and regions around the world.



Examples for Endangered Species

1. Giant Panda
2. Tiger
3. Asian Elephant
4. Snow Leopard
5. Blue Whale, etc.



Endemic Species: The endemic is derived from Greek. In Greek, "endemos" means "dwelling in a place". Indigenous, Native are the synonyms for the word endemic.

Endemic species are a group of species that are unique to a defined geographic location such as an island nation, country or other defined zone or habitat type.

Bicoloured frog (*Climotarsus curtipes*) is endemic to the Western Ghats of India.



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UNIT 3

NATURAL RESOURCES AND MANAGEMENT

3.1 Types of Natural Resources and Energy Resources

- 3.2 Water Conservation - Rain Water Harvesting: Watershed Management - Meaning and Importance.
- 3.3 Wasteland Reclamation, Soil Conservation, Afforestation - Meaning and Importance
- 3.4 Disaster - Definition and Types (Natural and Man-made), Self-protection during Disasters (Fire, Floods, Earthquakes)

3.1 TYPES OF NATURAL RESOURCES AND ENERGY RESOURCES

3.1(A) TYPES OF NATURAL RESOURCES

The word 'Resource' means 'a source of supply' or support generally held in reserve'. Living and non-living factors of nature used by humans for various purposes are called 'natural resources'. These are used to support human life and fulfil human requirements.

On the basis of origin, natural resources may be divided into two types. They are;

- 1) Biotic resources and 2) Abiotic resources

1. Biotic resources

The resources obtained from the biosphere are categorized as Biotic resources. The biosphere is made of living and organic material. As a result, the resources obtained from forests and animals are *living biotic resources* and the materials formed from decayed organic matter such as coal and petroleum categorized as "*fossil fuels*" are the biotic resources obtained from organic material.

2. Abiotic resources

The resources obtained from non-living, non-organic material are known as abiotic natural resources. It includes the resources such as land,

NATURAL RESOURCES AND MANAGEMENT

air, fresh water, earth metals and heavy metals including ores such as gold, iron, copper silver etc.

On the basis of stage of development, natural resources are further classified in the following ways.

- 1) Potential resources
- 2) Actual resources
- 3) Reserve resources and
- 4) Stock resources

1. Potential Resources

The resources that remain in the earth's crust and may be used in the future are referred to as potential resources.

Eg:- Petroleum

2. Actual Resources

The resources that have been surveyed, quantified, qualified and are currently being used for various purposes using technology are the actual resources.

Eg:- Wood, Wind etc

3. Reserve Resources

The part of an actual resource that can be developed profitably in the future is reserve resource.

Eg:- Wind

4. Stock Resources

These are the resources that have been surveyed, but cannot be used due to lack technology are the stock resources

Eg:- Gases such as hydrogen, sea water etc

3.1(B) ENERGY RESOURCES

Energy resources are all forms of fuels used in the modern world, either for heating, generation of electrical energy, or for other forms of energy conversion processes.

There are 10 main different sources of energy that are used in the world to generate power. They are;

- 1) Solar Energy
- 2) Wind Energy
- 3) Geothermal Energy
- 4) Hydrogen Energy
- 5) Tidal Energy
- 6) Wave Energy
- 7) Hydro power energy
- 8) Biomass Energy
- 9) Nuclear Power
- 10) Fossil fuels (coal, oil & Natural Gas)

Classification of Energy Resources

The energy resources can be roughly classified into three categories. They are:

- 1) Renewable energy resources
- 2) Fossil energy resources and
- 3) Nuclear energy resources

1. Renewable Energy Resources

The forms of energy that are naturally replenished on our planet are renewable energy resources.

Eg: Hydropower, Biomass, Wind, Wave, Tidal, Solar, Geothermal energy etc

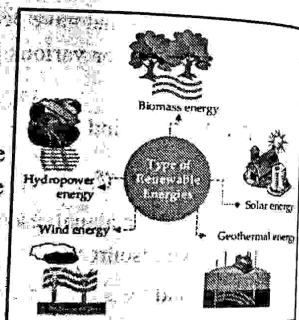
2. Fossil Energy Resources

The forms of energy resources obtained from dead plant and animal deposits created over the long history of planet Earth are fossil energy resources.

Eg: Fossil fuels - Coal, Oil and Natural gas. This energy resource is limited in supply and can get exhausted on continuous use i.e. "Non-renewable" in nature.

3. Nuclear Energy Resources

Deposits of certain radioactive elements present in Earth's crust are classified as nuclear energy resources. These resources are used as fuel for



nuclear fission-based power plants to produce nuclear energy. In this type of energy, using a small amount of raw material, large quantity of energy is produced. Even though, radioactive elements are limited in earth's crust and are non-renewable, the energy sources will last long for a long time. So it is also known as sustainable energy resource.

Further, these energy resources can be categorised into two types: Conventional sources of energy and Non conventional sources of energy.

The energy sources which have been in common use are called conventional sources of energy. Electricity, petrol, coal, wood, gas etc. are examples of conventional sources of energy.

The energy sources which are either new or have not been in common use are non-conventional sources of energy. Solar energy, wind energy, wave energy, geothermal energy, Biogas energy etc. are examples of non-conventional sources of energy.

3.2 WATER CONSERVATION-RAIN WATER HARVESTING

Water shed management - Meaning and importance.

WATER CONSERVATION

Water is an integral component of ecosystem. It is vital for the survival of plants and animals. Various human activities need water. With rapid urbanization and industrialization, acute shortage of water is observed. This is due to the over exploitation of surface water and rapid decline of ground water. This necessitates the need to *renew the water conservation methods followed by our ancestors and to invent innovative methods to conserve water*. The purpose of water conservation is *to recharge the depleting ground water and collect and use the rain water for various purposes*.

These objectives are fulfilled in the following two methods of water conservation.

3.2.1(A) Rain Water Harvesting

It is seen that in urban areas complete concretization of land in the form of buildings, roads, footpath, etc., has left too little earth exposed for rainwater to soak in. Thus, the water flows into the drains and moves into the natural water bodies [rivers / lake or sea] or reservoir. If this water were

1. Energy obtained from the temperature of the earth's crust.

to stay longer on the land, it would seep into ground and recharge ground water. This is becoming popular method of water conservation. Rain water harvesting is the conservation technique of collecting rain water and storing it in tanks for future use or putting it back to soil to recharge the ground water.

Techniques of Rainwater Harvesting

There are two main techniques of rain water harvesting:

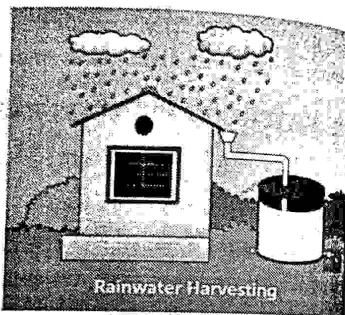
1. Surface rain water harvesting
2. Rain water harvesting to recharge the ground water.

1. Surface Rain Water Harvesting

In this method of harvesting, the roof is used as the catchment area and surface or ground water tanks are used as the storage area.

Steps Involved in Surface Rain Water Harvesting

1. As roof is used as the catchment area, roof surface must be kept clean.
2. The first rain water is not harvested as it is likely to contain suspended material.
3. Once the roof gets cleaned, the water from the roof is directed into primary filtration unit. This unit consists of a wire mesh to remove the unwanted suspended materials.
4. The water from primary filtration unit is directed into secondary filtration unit. In this unit, chemical treatment of water is done with potassium permanganate or chlorine. This purifies the water.
5. The filtered and purified water is then stored in the store tanks.
6. The storage tanks should not be directly exposed to sunlight to prevent the growth of algae.
7. Depending on the storage capacity, this water can be used for days or months.



2. Rain Water Harvesting to Recharge Ground Water

In this method of rain water harvesting, the water collected from the roofs or rain water flowing on the ground is directed into pits, dug wells, trenches, etc., to increase the ground water table.

1. Recharge pits of 1 - 2 m wide and 3 m deep are dug and filled with boulders, gravel and coarse sand to filter water.
2. Trenches of 0.5 - 1 m wide and 1 - 1.5 m deep and 10 - 20 m long are made and filling it with the filling material.
3. The dug wells are used into which run off water can be channelized but only after passing through filters.

3.2.1(B) Rain Water Harvesting-Importance

The importance of Rain water harvesting lies in the fact that it shows following advantages;

- a) Rain water harvesting helps the environment by reducing flooding and soil erosion.
- b) Rain water harvesting reduces the dependence on corporation municipal water and there by reduces the water bills.
- c) Rain water harvesting reduces the demand on ground water
- d) Rain harvested water can be used for non drinking purposes as well
- e) Rain harvested water can be used for plant growth purpose as well.

3.2.2(A) Water Shed Management

Watershed region is any area of land that catches rain water and drains into a stream, river or lake or allows it to seep into the ground. OR

Watershed management can be defined as 'a well defined area of land from which rain water drains out to a common point.

Water shed can occur in all shapes and sizes. The water shed that drains into rivers measures thousands of hectares whereas the watershed that drains into a pond or pool may measure only a few hectares.

Therefore, watershed management refers to the process of creating and implementing plans, programs and projects to sustain and enhance watershed functions that affect the plant, animal and human communities within a watershed boundary.

The main purpose of watershed management is to protect, maintain and restore water quality in our watersheds through integrated and holistic efforts.

3.2.2(B) Water Shed Management - Importance

The following are the importance of watershed management

- Water shed management mitigates droughts and floods.
- It increases the production and income of the water shed community
- It can prevent future community water shortages and poor water quality
- It increases the life of the downstream dams and reservoirs.
- It helps in recharging ground water levels.

3.3. WASTELAND RECLAMATION, SOIL CONSERVATION, AFFORESTATION-MEANING AND IMPORTANCE

3.3(A) Wasteland Reclamation - Meaning and Importance

Land is an important resource as it is put to different uses by man. Loss of vegetation cover leads to the loss of soil through erosion leading to the creation of wastelands.

Wastelands are lands which are unproductive, unfit for cultivation, grazing and other economic uses due to rough terrain and eroded soil. The lands which are waterlogged and saline are also termed as wastelands. The loss of soil fertility followed by erosion also leads to the conversion of marginal forest lands into waste lands.

If this goes on unchecked, it will affect the remaining land. So, conservation of soil, protecting the existing cultivable land and reclaiming the already depleted waste land assumes significance.

It is reported that every year about 3 million hectares of cropland are ruined by erosion and about 8 million hectares are taken away for non-agricultural purpose (plots for houses, roads, industries, reservoirs etc)

Wastelands are classified into two types;

- 1. Barren and uncultivable wasteland:** These lands cannot be brought under cultivation or economic use except at a very high cost Eg:- Sandy deserts, Gully land, stones/ leached land on hilly slopes, rocky lands etc.

- 2. Cultivable Wastelands:** These are the lands that are not cultivated for five years or more. It consists of land available for cultivation, but not used for cultivation.

Reclamation of Wastelands

Reclamation of wasteland means "re-claiming the land or use of land for productive purposes

Following are the methods of wasteland reclamation:

- 1) Afforestation:** It consists of growing a forest over cultivable wasteland.
- 2) Reforestation:** Growing the forest again over the land where they are once existed and were destroyed due to overgrazing, fires, excessive tree cutting etc.
- 3) Providing Surface Cover:** This is the easiest way to protect the land surface from soil erosion. Here crop residue is left on the land after harvesting. This prevents the land surface from soil erosion.
- 4) Mulching:** Here protective cover of organic matter and stalks of plants (Cotton, Tobacco, etc.) are used, to reduce evaporation and thereby helping in retaining soil moisture & reduce soil erosion.
- 5) Changing Ground Topography on Downhill:** Running water erodes the hill soil and carries the soil along with it. This can be minimized by altering the ground topography by adapting;
 - a) Strip farming**
 - b) Terracing**
 - c) Contour ploughing, etc.**
- 6) Leaching:** In salt affected land, the salinity can be minimized by leaching them with more water
- 7) Changing Agricultural Practices:** Adapting mixed Cropping & Crop rotation, soil fertility can be improved.
- 8) Ecological Succession:** It consists of natural development or re development of an ecosystem which helps in reclaiming the minerally deficient soil of wasteland.

Importance/Need for Wasteland Reclamation

1. It makes the soil fertile by preventing soil erosion and conserving the moisture of soil.
2. It helps in maintaining the ecological balance in the area/region.
3. It helps in increasing the forest cover and thereby is responsible for maintaining the local climatic condition.
4. It ensures a constant supply of fuel, fodder and timber for local use.
5. It provides a source of income for the rural poor.

3.3(B) SOIL CONSERVATION: MEANING AND IMPORTANCE

Soil conservation is a combination of practices used to protect the soil from degradation.

Importance/Need for Soil Conservation

1. To Maintain Adequate Amount of Organic Matter and Biological Life in the Soil: Organic matter and biological life present in soil accounts for 90-95% of the total soil productivity

2. To Ensure a Secure Food Supply at Reasonable Price: Soil conservation has proven to increase the quality and quantity of crop yield over a long term as it keeps the fertile top soil in its place and thereby preserves the long term productivity of the soil.

3. To grow enough food not only for ourselves but also for those countries where there is shortage of food grains.

4. To Improve Water Quality: Agriculture and urban soil erosion are major sources of sedimentation and contamination of water supplies. Soil conservation prevents this.

5. To Improve Wildlife Habitat: Soil conservation practices such as providing buffer strips and wind breaks or replacing the soil organic matter greatly enhances the quality of environment for wild life.

6. To help to create an environment free of pollution where we can live safely.

7. For the future of our next generation, so that they will have enough soil to support life.

3.3(C) AFFORESTATION: MEANING AND IMPORTANCE

"The forest is not a resource for us. It is life itself. It is the only place for us to live".

- Evaristo Nzinguag Ikana

Afforestation is the process of planting trees in areas devoid of any forest cover i.e Afforestation "is the establishment of a forest or stand of trees in an area where there was no previous tree cover".

Importance of Afforestation

The process of Afforestation holds the following importance/ advantage as it is the process of creating new forest. They are;

1. Afforestation Provides an Alternate Source of Tree Products:

The rate at which trees grow naturally in forests is much slower than the rate at which the trees in forests are being cut down for various products of human use. Increased demand for tree products puts pressure on forests, resulting in deforestation. This is taken care now as afforestation serves the purpose of being alternate source of tree products for human use.

2. Afforestation has Increased Supply of Trees that are in High Demand: It is observed that there is demand for specific types of trees for commercial purposes. Afforestation allows the stake holders to plant the type of trees in demand, promoting the fast propagation of specific types of trees.

3. Protection of Natural Forests: Afforestation is making the protection of natural forests a possibility now as alternate source of tree products are easily and readily available. Along with this, protecting natural forests is leading to other benefits such as;

- a) Preservation of catchments
- b) Preservation of wetlands and
- c) Preservation of riverside zones

4. Afforestation Leads to Environmental Benefits: Planting trees in a barren land helps to create a new ecosystem. Due to this reason, majority of governments use afforestation to revert the arid and semi-arid areas to productive areas.

5. Afforestation Brings About Value Addition to a Region: Easiest way to transform a barren land is by planting trees. Land with trees is always